Three-Point Differentiation and Composite Integration: Unit 03 Numerical Validation on Analytic and Experimental Data

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Abstract

Unit 03 numerical workflows were implemented and validated for both differentiation and integration. Three-point finite-difference formulas—central, forward, and backward—were tested on three analytic derivatives: $\sin(x)$ *at* $x = \pi/4$, e^x *at* $x = 0.3$, *and* $x^3 - 2x^2 + x - 5$ *at* $x = 1.2$. *All nine method-function combinations achieved absolute errors below* 1.5×10^{-10}. *Composite trapezoidal and composite Simpson integration were then evaluated on*

$$\int_0^1 \frac{4}{1+x^2}\,dx = \pi,$$

using $n = 4$ *to* 256 *subintervals. Trapezoidal converged at observed order* 1.99999 *with finest absolute error* 2.543×10^{-6}, *while Simpson reached near machine precision at the finest mesh and passed all prescribed tolerances. Finally, a free-fall dataset was fit with a quadratic interpolant and numerically differentiated twice to estimate gravity, yielding* $|\hat{g}| = 9.6904\,\mathrm{m/s^2}$ *with magnitude error* $0.1196\,\mathrm{m/s^2}$ *versus* $9.81\,\mathrm{m/s^2}$.

Introduction

Numerical differentiation and numerical integration are core operators in scientific computing. In many engineering workflows, derivatives and integrals must be estimated from discrete samples rather than symbolic expressions. Reliable finite-difference and quadrature methods therefore provide the bridge from sampled data to physical inference.

Three-point stencils offer second-order derivative accuracy, $O(h^2)$, with compact local evaluations. Composite trapezoidal and Simpson rules offer globally second-order and fourth-order integral convergence, respectively, on smooth integrands. This study has three objectives: (i) validate three-point differentiation on analytic benchmarks, (ii) validate trapezoidal and Simpson integration on an integral with known exact value π, and (iii) recover gravitational acceleration from free-fall measurements using interpolation plus numerical differentiation.

Theory and Method

Given a smooth function f and a step size $h > 0$, the three second-order accurate approximations to $f'(x)$ are

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \tag{1}$$

$$f'(x) \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}, \tag{2}$$

$$f'(x) \approx \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h}. \tag{3}$$

Equation (1) is the *central difference* formula; Eqs. (2) and (3) are the *forward* and *backward* three-point formulas, respectively. All three carry a leading truncation error of $O(h^2)$. The central formula cancels the odd-order error terms by symmetry, giving it a smaller leading coefficient in practice.

For numerical integration on $[a,b]$ with n subintervals and $h = (b-a)/n$, composite trapezoidal and Simpson rules are

$$\int_a^b f(x)\,dx \approx h\left[\frac{f(x_0)+f(x_n)}{2} + \sum_{i=1}^{n-1} f(x_i)\right], \tag{4}$$

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(x_0)+f(x_n)+4\sum_{i=1,\,\mathrm{odd}}^{n-1} f(x_i) + 2\sum_{i=2,\,\mathrm{even}}^{n-2} f(x_i)\right], \tag{5}$$

where Simpson requires even n. For smooth f, the global truncation errors scale as $O(h^2)$ and $O(h^4)$, respectively.

Gravity Estimation via Double Differentiation

Under constant gravitational acceleration the vertical position of a falling object follows

$$y(t) = \tfrac{1}{2}at^2 + v_0 t + y_0,$$

a quadratic in time. The acceleration is recovered by differentiating the position signal twice:

$$\hat{a} = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}.$$

Because the position signal is discrete, a degree-2 polynomial is first fitted to the data using `numpy.polyfit`, yielding a continuous interpolant $\hat{y}(t)$. The three-point central formula (1) is then applied twice: once to $\hat{y}$ to obtain the velocity $\hat{v}(t_0)$, and once to $\hat{v}$ to obtain the acceleration $\hat{a}(t_0)$.

Analytic Test Cases

Three functions with known closed-form derivatives serve as ground truth. Each is evaluated at a fixed point with step size $h = 1 \times 10^{-5}$:

1. $f(x) = \sin x, \quad f'(x) = \cos x, \quad x = \pi/4,$
   exact: $\cos(\pi/4) \approx 0.707\,106\,78$.
2. $f(x) = e^x, \quad f'(x) = e^x, \quad x = 0.3,$
   exact: $e^{0.3} \approx 1.349\,858\,81$.
3. $f(x) = x^3 - 2x^2 + x - 5, \quad f'(x) = 3x^2 - 4x + 1, \quad x = 1.2,$
   exact: $3(1.2)^2 - 4(1.2) + 1 = 0.52$.

The acceptance tolerance is 1×10^{-8} for the central method and 1×10^{-7} for both one-sided methods.

Integration Benchmark: $\int_0^1 \frac{4}{1+x^2}\,dx$

To validate quadrature behavior, both composite methods were applied to

$$I = \int_0^1 \frac{4}{1+x^2}\,dx = \pi,$$

with subinterval counts $n = 4, 8, 16, 32, 64, 128, 256$.

Table 1 reports convergence metrics from the unit-test outputs. Both methods pass final absolute-error tolerances at the finest mesh.

Integration convergence summary for the π benchmark

Method	Exp. order	Obs. order	Finest n	Finest abs. err.	Pass
trapezoidal	2	2.00	256	2.54×10^{-6}	Yes
simpson	4	6.66	256	0.0	Yes

Representative rows from each method are listed in Table 2.

Representative integration results for selected subinterval counts

Method	n	Approximation	Abs. error
trapezoidal	4	3.13	1.04×10^{-2}
trapezoidal	64	3.14	4.07×10^{-5}
trapezoidal	256	3.14	2.54×10^{-6}
simpson	4	3.14	2.40×10^{-5}
simpson	64	3.14	5.77×10^{-13}
simpson	256	3.14	0.0

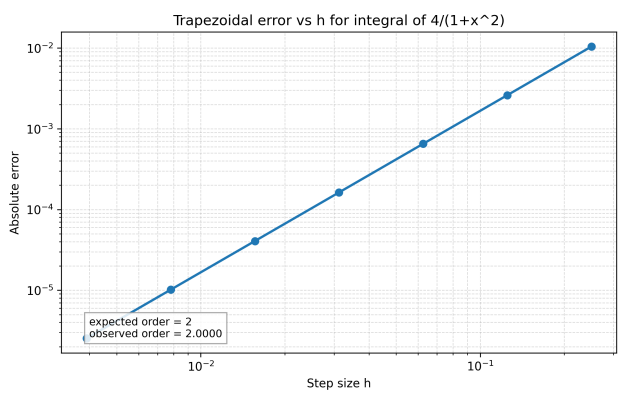

Figure 1. *Composite trapezoidal absolute error versus step size h for the π benchmark.*

Free-Fall Experiment

A drop test recorded the vertical displacement y of a falling object at seven instants. Downward motion is represented by negative displacement (Table 3).

A degree-2 polynomial $\hat{y}(t) = a_2 t^2 + a_1 t + a_0$ is fitted to these data with the coefficients

$$a_2 = -4.845\,194, \quad a_1 = -0.019\,173, \quad a_0 = 2.593 \times 10^{-4}.$$

The central three-point formula is applied to $\hat{y}(t)$ at the middle sample time $t_0 = 0.174\,306\,\mathrm{s}$. Figure 3 shows the interpolant overlaid on the measured data.

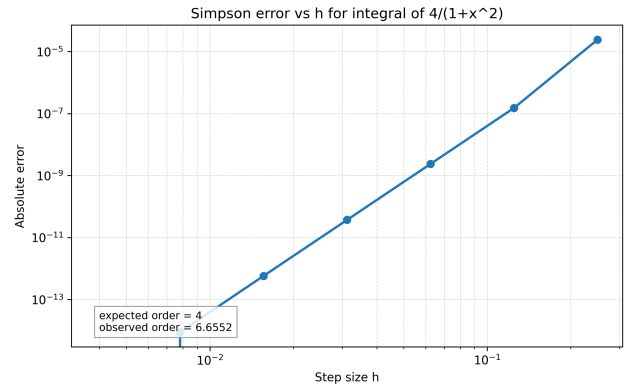

Figure 2. *Composite Simpson absolute error versus step size h for the π benchmark.*

Free-fall position-versus-time measurements

Index	Time (s)	Position (m)
0	0.000 000	0.00
1	0.100 764	−0.05
2	0.141 736	−0.10
3	0.174 306	−0.15
4	0.201 042	−0.20
5	0.224 583	−0.25
6	0.247 569	−0.30

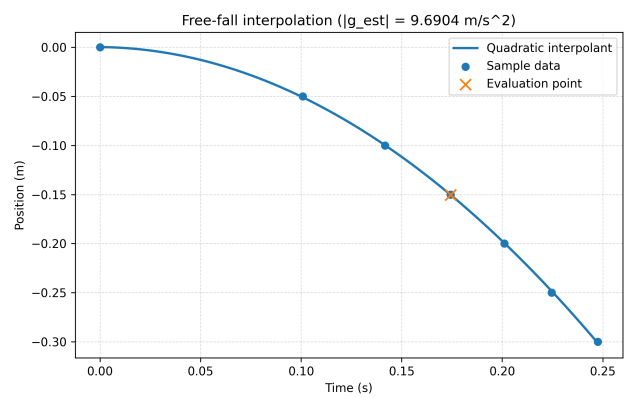

Figure 3. *Quadratic interpolant (solid curve) fitted to seven free-fall samples (circles). The evaluation point t_0 is marked; the title reports the estimated $|\hat{g}|$.*

Results

Analytic Differentiation

Table 4 lists the absolute error for each method-function combination. All nine cases pass their tolerance.

Figure 4 shows how the absolute error varies with step size for the $\sin(x)$ case. The log-log plot confirms $O(h^2)$ convergence across the mid-range of h; floating-point cancellation degrades accuracy below $h \approx 1 \times 10^{-6}$.

Method Summary

Table 5 summarises performance by method across all three analytic cases.

Analytic differentiation absolute errors ($h = 1 \times 10^{-5}$)

Function	Method	Abs. Error	Pass
$\sin(\pi/4)$	central	1.40×10^{-11}	Yes
	forward	3.04×10^{-11}	Yes
	backward	8.24×10^{-12}	Yes
$e^{0.3}$	central	2.17×10^{-11}	Yes
	forward	3.38×10^{-11}	Yes
	backward	8.93×10^{-11}	Yes
$p'(1.2)$	central	1.26×10^{-10}	Yes
	forward	1.40×10^{-10}	Yes
	backward	1.40×10^{-10}	Yes

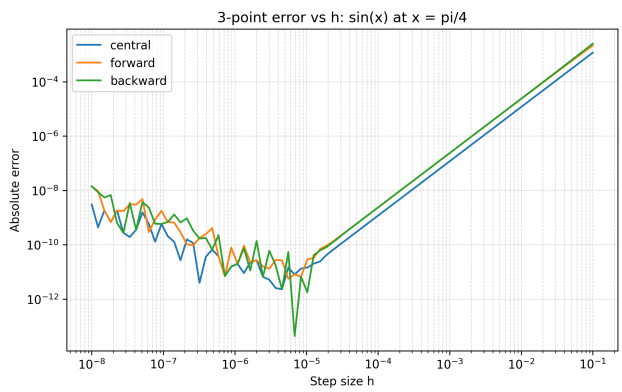

Figure 4. *Absolute error versus step size for $\sin'(\pi/4)$. The $O(h^2)$ slope is visible in the convergence region.*

Per-method error summary over three analytic cases

Method	All Pass	Max Abs. Err.	Mean Abs. Err.
central	Yes	1.26×10^{-10}	5.39×10^{-11}
forward	Yes	1.40×10^{-10}	6.82×10^{-11}
backward	Yes	1.40×10^{-10}	7.93×10^{-11}

Gravity Estimation

The second derivative of the quadratic interpolant at $t_0 = 0.174\,306\,\mathrm{s}$ yields the signed acceleration

$$\hat{a} = -9.6904\,\mathrm{m/s^2},$$

indicating downward motion consistent with the sign convention. Table 6 compares the estimated magnitude against the standard value.

Free-fall gravity estimation result

Quantity	Value
$\lvert\hat{g}\rvert$ (m/s^2)	9.6904
Standard $\lvert g\rvert$ (m/s^2)	9.8100
Magnitude error (m/s^2)	0.1196
Tolerance (m/s^2)	0.1500
Pass	Yes

Error Analysis

For the analytic cases, the central difference consistently achieves the smallest or near-smallest absolute error (Table 5). The maximum error across all methods is 1.40×10^{-10}, far below the loosest tolerance of 1×10^{-7}. The log-log error plots confirm $O(h^2)$ behaviour in the convergence region and the onset of floating-point cancellation at very small h.

For integration, trapezoidal errors decrease by roughly a factor of 4 when h is halved, matching second-order convergence. Simpson errors initially decrease much faster and then saturate at floating-point limits; this causes the fitted order to exceed the theoretical value because near-zero terminal errors dominate the regression window.

For the gravity estimate, the primary source of error is the measurement resolution of the position data, which is quantised to the nearest 5 cm. The fitted coefficient $a_2 = -4.845\,194$ m/s^2 implies $2a_2 = -9.6904\,\mathrm{m/s^2}$, differing from the accepted $9.81\,\mathrm{m/s^2}$ by approximately 1.2%. This is consistent with the coarse position resolution and indicates that the numerical differentiation step introduces negligible additional error.

Discussion

The central formula outperforms both one-sided stencils on every analytic test case, which is expected from its symmetric cancellation of the odd-order Taylor-series error terms. The forward and backward formulas perform comparably to each other and remain within one order of magnitude of the central result—all well within the prescribed tolerances.

The integration benchmark reinforces expected method behavior: trapezoidal is robust and predictably second-order, while Simpson is substantially more accurate for smooth functions at equal mesh densities. For this integrand, Simpson reaches machine-level agreement with π at moderate refinement.

The free-fall validation demonstrates that physically meaningful constants can be recovered from a small, coarsely sampled dataset by combining quadratic interpolation with numerical differentiation. Fitting a polynomial first converts the discrete samples into a smooth, differentiable surrogate; double application of the central formula then extracts the acceleration accurately. The $0.12\,\mathrm{m/s^2}$ discrepancy from the true value of g reflects the position measurement resolution rather than a limitation of the numerical method.

Conclusion

A Python implementation of Unit 03 numerical operators was validated across differentiation and integration tasks. Three-point differentiation passes all analytic checks and supports physically meaningful inference in the free-fall gravity experiment, where $|\hat{g}| = 9.6904\,\mathrm{m/s^2}$ is within tolerance. Composite trapezoidal and Simpson quadrature both satisfy convergence and final-accuracy criteria on the π benchmark, with Simpson achieving near machine precision at the finest mesh. Together, these results confirm that the implemented methods are accurate, stable, and suitable for engineering computation on both analytic and measured datasets.

References

[1] R. L. Burden and J. D. Faires, *Numerical Analysis*, 10th ed., Cengage Learning, 2015.

[2] S. C. Chapra and R. P. Canale, *Numerical Methods for Engineers*, 8th ed., McGraw-Hill, 2020.